Economics and Computation (2026 Fall)

Quiz 2

Part I: Cake Cutting

Throughout Part I, the cake is $[0, 1]$, and valuations are nonnegative, additive, and divisible, with $V_i([0, 1]) = 1$. A density f_i represents $V_i(X) = \int_X f_i(x)\, dx$. Individual endpoints have value zero.

Problem 1. (25%) **Dubins–Spanier on a small instance.**

There are three agents. Their densities are constant on each of the following intervals:

	$[0, \frac{1}{3}]$	$[\frac{1}{3}, \frac{2}{3}]$	$[\frac{2}{3}, 1]$
f_1	3	0	0
f_2	0	2	1
f_3	2	0	1

Use the discrete Dubins–Spanier procedure from the lecture. At a stage with remaining cake $[x, 1]$, each active agent i marks the leftmost point y_i satisfying $V_i([x, y_i]) = \frac{1}{3}$. The agent with the leftmost mark receives that prefix and leaves. The final agent receives all remaining cake. Break ties between agents by smaller index.

(a) (15%) Calculate every active agent's mark at each stage. Give the three final pieces A_1, A_2, A_3.

(b) (10%) Compute the 3×3 table whose (i, j) entry is $V_i(A_j)$. Is the allocation proportional? Is it envy-free? Identify every ordered pair (i, j) for which agent i envies agent j.

Problem 2. (20%) **Optimal proportional cake cutting.** There are two agents. Their densities on $L = [0, \frac{1}{2}]$ and $R = [\frac{1}{2}, 1]$ are

	L	R
f_1	2	0
f_2	$\frac{3}{2}$	$\frac{1}{2}$

Let x and y be the fractions of the *lengths* of L and R, respectively, allocated to agent 1. Agent 2 receives the remaining fractions. Write $\mathrm{SW}(A) = V_1(A_1) + V_2(A_2)$.

(a) (20%) Express both agents' utilities in terms of x, y. Formulate a linear program maximizing the utilitarian social welfare (USW) subject to proportionality and $0 \le x, y \le 1$.

Part II: Fair Allocation of Indivisible Goods

Write $v_i(S) = \sum_{g \in S} v_i(g)$. Use the lecture's EFX convention: the required inequality must hold after removing *every* good from the other agent's bundle.

Problem 3. (30%) **Checking EFX.**

Two agents have the following values:

	g_1	g_2	g_3	g_4
v_1	4	5	2	1
v_2	2	4	3	2

Consider the allocation "$A = (\{g_1\}, \{g_2, g_3, g_4\})$", where the first bundle in the pair belongs to agent 1.

(a) (10%) Is A an EFX allocation? (Just answer "yes" or "no")

(b) (20%) Find the largest $\alpha \in [0, 1]$ for which A is α-EFX, meaning $v_i(A_i) \geq \alpha\, v_i(A_j \setminus \{g\})$ for every i, j and every $g \in A_j$.

Problem 4. (25%) **Computing and attaining maximin shares.**

Two agents value three goods as follows:

	g_1	g_2	g_3
v_1	8	7	6
v_2	7	8	5

Recall
$$\mu_i^2(M) = \max_{(P_1,P_2) \text{ partition of } M} \min_{k \in \{1,2\}} v_i(P_k).$$

An MMS allocation gives every agent i value at least $\mu_i^2(M)$.

(a) (15%) Compute $\mu_i^2(M)$ for each agent. For each value, give a partition attaining it and prove a matching upper bound.

(b) (10%) Find a complete MMS allocation. List all bundles and verify the two MMS guarantees.

Problem 5. (20%) **The p-mean and Nash social welfare.**

Let $u_i > 0$ be agent i's utility, for $i = 1, 2, \ldots, n$; for example, $u_i = v_i(A_i)$ for an allocation $A = (A_1, A_2, \ldots, A_n)$. For every $p \in \mathbb{R} \setminus \{0\}$, define the p-mean welfare by

$$W_p = \left(\frac{1}{n} \sum_{i=1}^{n} u_i^p \right)^{1/p}.$$

Prove that W_p converges to the normalized Nash social welfare as $p \to 0$; that is, show that

$$\lim_{p \to 0} W_p = \left(\prod_{i=1}^{n} u_i \right)^{1/n}.$$